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Optimal Design of a Local Energy Network

Case study of Marpha Community,

Mustang District, Nepal

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1. Introduction

Marpha is a high-altitude village located in the Mustang District of the Gandaki Province, Nepal, situated at an elevation of approximately 2,670 meters above sea level. Nestled within the rain shadow of the Annapurna and Dhaulagiri massifs, the region is characterized by a semi-arid, alpine climate with extreme seasonal temperature fluctuations. Historically known for its "Marpha Apple" orchards and as a vital node on the Annapurna Circuit trekking route, the settlement functions as both a hub for agricultural production and a significant center for the regional tourism industry [1].

High-altitude mountain communities are among the most energy-vulnerable settlements, requiring reliable and resilient power sources to sustain essential livelihoods and agricultural productivity. With rising fuel costs for transportation and global sustainability mandates, integrating decentralized renewable energy solutions such as solar photovoltaic systems and battery storage has become a vital strategy to reduce dependence on the fragile national grid and enhance regional energy autonomy. This study examines the feasibility of a decentralized solar photovoltaic microgrid in Marpha, Mustang, leveraging local environmental conditions to ensure uninterrupted power while creating a sustainable model for community-led energy management.

A key characteristic of Marpha's energy profile is its extreme seasonal and diurnal demand variation. Due to the remote nature of the settlement and the specific energy needs of its primary industries, the community's energy profile was modelled based on a mix of residential demand. The estimated annual consumption is **1.45 GWh**, with an equivalent energy cost of approximately **NPR 18.2 million**, assuming current market rates for grid electricity and supplemental diesel fuel [2].

A battery energy storage system (BESS) has been incorporated into the design to maximize system stability and bridge the gap between generation peaks and nighttime demand. A minimum State of Charge (SOC) of **20%** was selected as the baseline to balance battery longevity with the need for a high emergency reserve in a region prone to natural isolation, including the analysis of such minimum SOC levels on the behaviour of the system.

By modelling system performance based on high-resolution meteorological data and local load profiles, this study takes a data-driven approach to optimally sizing a PV-Battery system that minimizes reliance on external fossil fuels and unstable transmission lines.

Ultimately, this study underscores how a decentralized hybrid energy integration and strategic storage management can transform the economic landscape of mountain communities like Marpha. Through optimized local resource utilization and smart energy management, Marpha can transition from an energy-dependent settlement into a self-sufficient model of rural Himalayan resilience, paving the way for a greener, more secure future for Nepal's high-altitude regions.

2. Case Study

Mustang District is in Gandaki Province in northern Nepal, within the high Himalayan region, and extends northward to the Tibetan Plateau. It is one of the most remote and sparsely populated districts in the country, covering an area of about 3,573 km². Mustang experiences a cool, semi-arid trans-Himalayan climate as it lies in the rain-shadow area of the Annapurna range. The district has an average annual temperature of around 10.9°C and receives approximately 300 mm of precipitation annually. During the summer season, temperatures can exceed 26°C, while in winter they can fall to as low as -9°C. Situated at an altitude of approximately 2,670 meters[3]. This project addresses the energy needs of a rural cluster of 50 households that lack reliable access to the national grid.

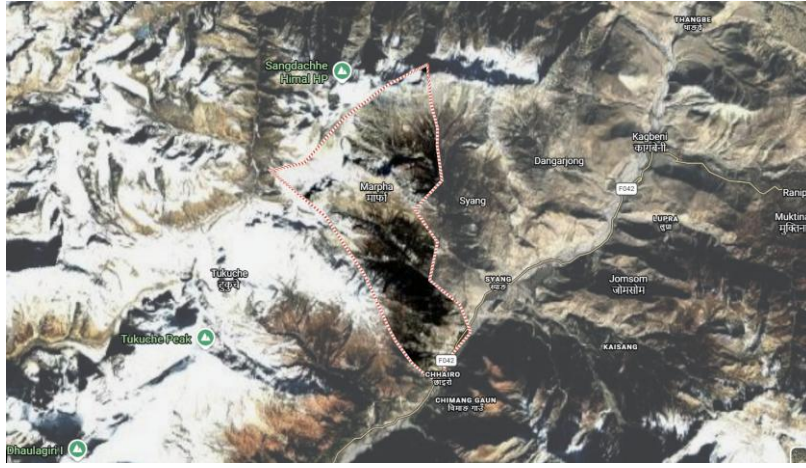


Figure 1. Marpha, Mustang District, Nepal (28.81N, 83.64E)

The consumption data and load profiles were modelled based on a typical high-altitude Himalayan household, where energy demand is heavily influenced by seasonal temperature extremes. The energy demand in high-altitude villages like Marpha is characterized by unique geographic challenges. While solar radiation is high (exceeding 300 days of sunshine annually), the ambient temperature often drops well below freezing, impacting battery performance and building thermal requirements.

The primary objectives of this energy optimization project are to implement a microgrid system using a Solar PV and a Battery Energy Storage System (BESS). The transition seeks to displace traditional fuels by reducing the dependency on firewood and expensive kerosene, provide an uninterrupted 24/7 power supply for lighting and communication, and improve public health by reducing the indoor air pollution caused by traditional cooking fires.

3. Data Collection and Manipulation

The energy system design for Marpha, Mustang District, Nepal, was based on three main input datasets which are solar irradiance, ambient temperature, and electrical load demand.

3.1. Solar Irradiance Data

The Solar Irradiance data was obtained from Meteonorm, which provides typical meteorological data for specified locations and can return hourly or monthly values for solar-energy applications.

For this study, the solar irradiance was processed on a monthly-hourly basis. The hourly values for each month were averaged across all days in that month. In practical terms, all values recorded at a given hour, such as 06:00, were averaged to obtain one representative 06:00 value for that month; the same procedure was repeated for all 24 hours and for all months. This produced a smoothed 24-hour average profile for each month for both irradiance and temperature. The advantage of this method is that it reduces the influence of short-term day-to-day fluctuations while preserving the typical daily trend needed for system sizing and simulation. Since the MATLAB model operates on fixed hourly time steps, this aggregation approach provided a consistent and computationally practical set of climatic inputs.

The irradiance results show a clear daytime profile for each month, with values rising after sunrise, peaking around midday, and returning to zero at night. This behaviour is physically consistent with solar resource availability. However, the plotted monthly curves do not show the simple pattern of “summer always higher than winter” that is often expected in lower-altitude locations. For Marpha, this is a defensible result.

An explanation comes from the broader solar-climate literature for Nepal. Studies of seasonal solar radiation in Nepal report that mean solar radiation is often higher in spring than in summer because monsoonal cloud cover reduces incoming radiation during the summer period [4]. Studies also note that mountainous regions can experience very high instantaneous solar radiation under clear-sky conditions because of reduced atmospheric attenuation at high altitude [5]. Therefore, higher hourly irradiance peaks

in some non-summer months do not necessarily indicate an error in the dataset; rather, they are consistent with the combined effects of altitude, mountain climate, and seasonal cloudiness.

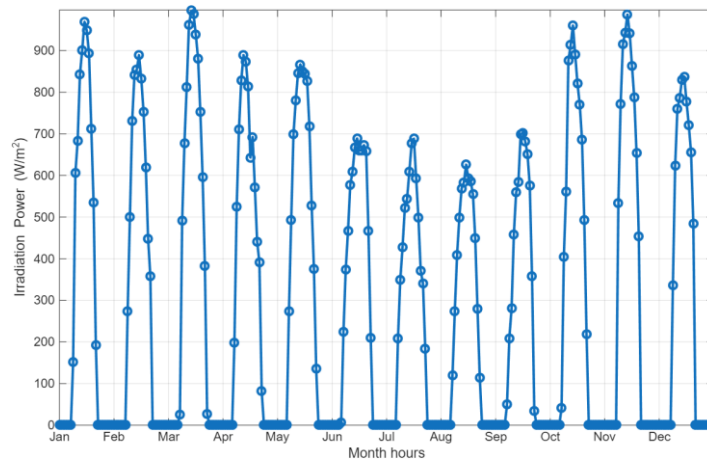


Figure 2. Annual Solar irradiation (source: Metronorm)

3.2. Ambient Temperature Data

The ambient temperature data was obtained from Meteonorm, which provides typical meteorological data for specified locations and can return hourly or monthly values for solar-energy applications.

For this study, the temperature data was processed on a monthly-hourly basis. The hourly values for each month were averaged across all days in that month. This produced a smoothed 24-hour average temperature profile for each month of the year.

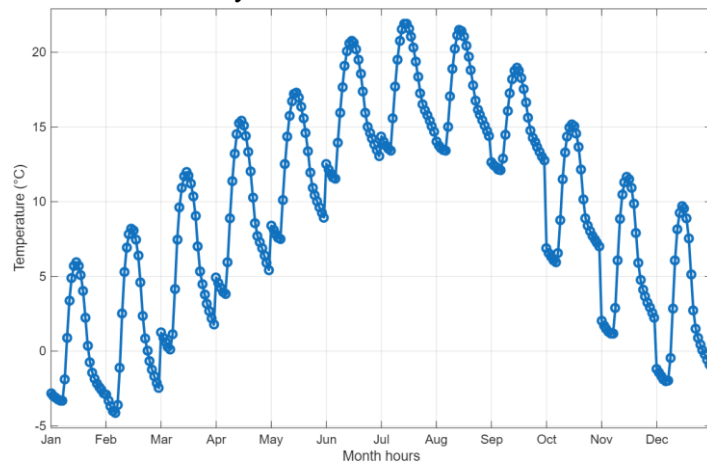


Figure 3. Annual Ambient Temperature (Source: Metronorm)

The temperature dataset showed an expected seasonal trend across the year, with lower values in winter and higher values in summer. In the winter months, especially December to February, the hourly average temperatures are low and, in some cases, fall below 0 °C during the night and early morning. As the year progresses into spring, the temperatures gradually rise, with both daytime and nighttime values becoming noticeably warmer. The highest temperatures occur in the summer period, particularly around June to August, when daytime values reach above 20 °C, while nighttime temperatures remain significantly milder than in winter. This creates a smooth annual cycle in which summer is clearly the warmest season and winter the coldest, even though the irradiance pattern does not follow the same trend. This temperature profile is also important for determine the heating demand for the load profile.

3.3. Electrical Load Profile

The load profile was developed from the community energy needs identified for Marpha, including lighting, cooking, television use, space heating, and water heating. These end uses were translated into an hourly demand profile suitable for the MATLAB sizing model. For this project, the electricity consumption data for the region was first manipulated and analysed using Python to determine the best approach for

studying consumption patterns. The manipulation was based on the number of households and annual consumption per household. Furthermore, the load share was divided based on the common household electrical appliances, such as lighting, television, cooking, house and water heating, and other shares. The literature reports that for these parts the lighting load share is highest [6].

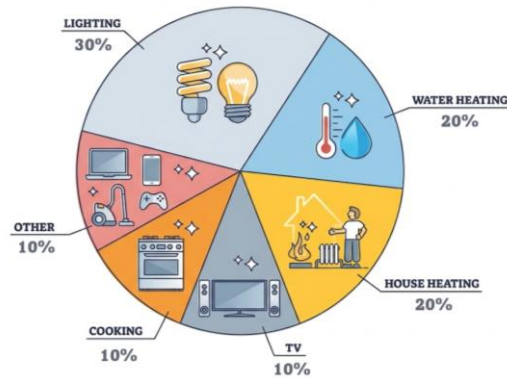


Figure 4. Load Shares (adapted from (Rijal et al., n.d.))

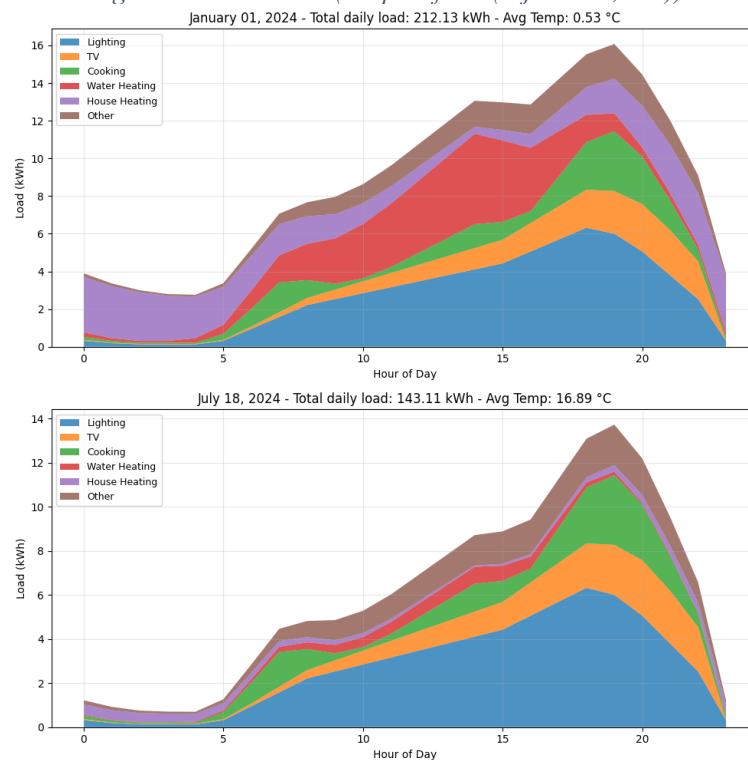


Figure 5. Load Simulation of two temperature scenario, 1st January 2024 (Winter) and 18th July 2024 (Summer)

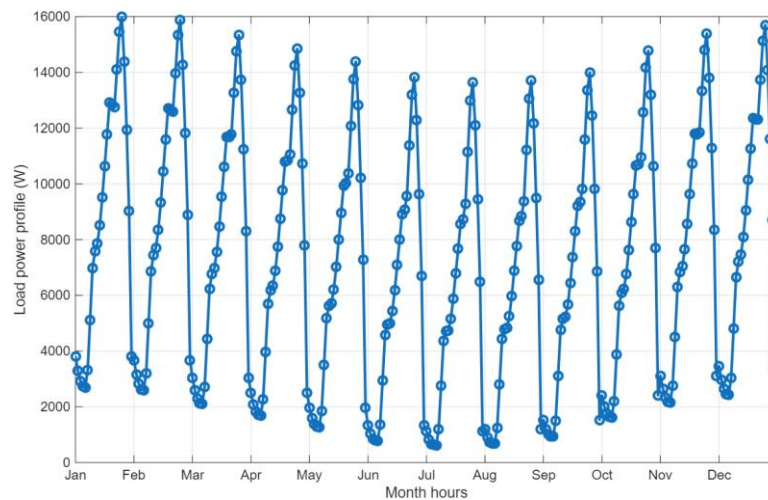


Figure 6. Load profile for Marpha

For the analysis, the data was averaged to represent hourly values for each month, ensuring key trends were retained while smoothing fluctuations to highlight seasonal variations. The data was then processed and cleaned in MATLAB, where duplicates were removed, missing values were handled, and the data was plotted for visual analysis. These refined insights formed the basis for configuring the microgrid control strategy.

This load profile, together with the processed solar irradiance and temperature data, formed the basis for the PV-battery system design. Overall, the data preparation process was aimed at producing realistic and consistent inputs for optimization. Meteonorm provided the climatic basis, monthly-hourly averaging produced stable profiles, and the resulting trends are typical for a high-altitude Himalayan location, like Marpha, influenced by complex terrain and seasonal monsoon dynamics.

4. System Design and Optimization

4.1. Proposed System Design

The proposed hybrid energy system integrates solar PV, and battery storage through a DC microgrid to reliably power the Marpha community. PV supply the load directly, while surplus energy charges the battery or is exported to the grid. If renewable supply is insufficient, the battery discharges to meet demand. A control strategy ensures efficient power flow, prioritizing load, optimizing battery use, and maximizing renewable utilization, all while supporting the livelihood of the community.

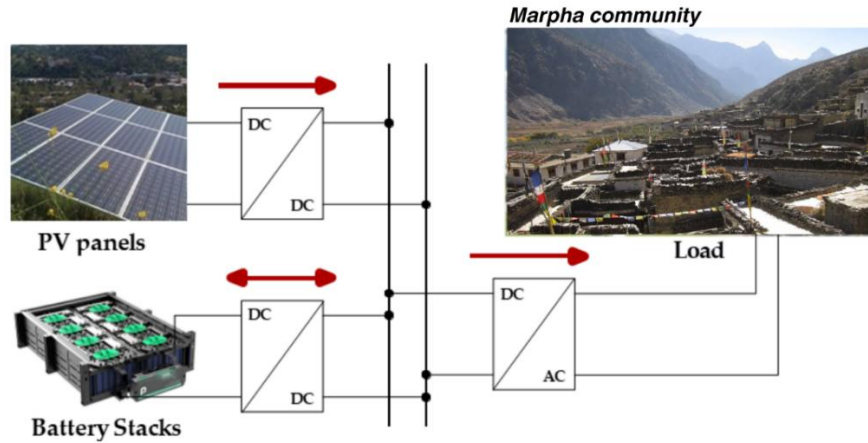


Figure 7. Proposed PV-system architecture for Marpha

4.2. Code Development

The MATLAB code was developed to simulate the system operation over an annual profile and determine a suitable configuration of solar panels and battery storage required to meet the electrical demand for the community. The code reads all the dataset provided and defines the main system parameters, including PV panel characteristics, battery cell capacity, converter efficiencies, battery efficiency, and cost coefficients. Based on the irradiance and temperature profiles, the code computes the PV power generation at each time step, while the load profile represents the community demand that must be supplied continuously. The code flow of the modelled system in MATLAB is as shown in *Figure 8*.

Power flow: A power balance routine is applied to describe how energy flows through the system. At every time step, the code compares the available PV power with the load demand. When PV generation is greater than the load, the excess energy is directed to battery charging if storage capacity is still available; otherwise, the extra energy is counted as lost energy. When PV generation is lower than the load, the battery discharges to cover the deficit if its state of charge remains above the minimum allowable limit. If both PV generation and battery storage are insufficient, the remaining deficit is recorded as unsupplied power, which is not applicable in our case, because all demand is met throughout the year.

Optimization: A genetic algorithm (GA) implemented via MATLAB's ga function was employed to solve the optimization problem. The decision variables included the number of PV panels (N_{PV}), battery cells (N_{cell}), wind turbines (N_{WT}), and the initial state of charge (SOC), each bounded within predefined limits. Due to the stochastic nature of GA, the optimization was performed ten times using a large population size of 10,000 and a maximum of 1,000 generations, while keeping other parameters at their default settings. All solutions were recorded, and the configuration yielding the minimum investment cost was selected as the final optimal design to ensure robustness and consistency of the results.

Proposed results: Once the optimal solution is obtained, the code outputs the results of the proposed system. These include the optimal number of PV panels, the optimal number of battery cells, the initial state of charge, the capital expenditure (CAPEX), the total lost energy, and the levelized cost of electricity (LCOE). It also generates plots of the power balance showing PV generation, battery power and lost power over the year which is shown in *Figure 11* and *Table 5*.

Solar PV Output Power

The Solar PV Power was calculated using the equation below:

$$P_{PV}(t) = N_{PV} * \eta_{PV} * P_{STC} * \frac{GSR(t)}{G_{STC}} * (1 - C_T(T(t) - T_{STC}))$$

Where,

N_{PV} = number of panels	G_{STC} = STC Global Solar Radiation (1 kW)
η_{PV} = panel electrical efficiency	C_T = temperature coefficient
P_{STC} = nominal power under Standard Test Conditions (STC)	T_{STC} = STC temperature (25°C)

Batteries

The number of battery necessary were calculated using the equation:

$$SOC(t + \Delta t) = SOC(t) - \eta_{Bat} * \frac{P_{Bat}}{N_{Cell} C_{Cell}} * \Delta t$$

Where,

SOC : State of charge	N_{Cell} : Number of cells
η_{Bat} : Battery efficiency	C_{Cell} : Cell capacity
P_{Bat} : Battery power	

LCOE

$$C_{lifetime} = C_{total} + \sum_{t=1}^T \frac{C_{total} \cdot M}{(1 + r)^t}$$

where:

M is the maintenance cost fraction (2.5% of total capital investment),

T is the system lifetime (25 years).

R=8% is the discount rate

$$Levelised\ Cost\ of\ Electricity\ (LCOE) = \frac{C_{life_time}}{E_{life_time}}$$

4.3. Technical Specification

The technical specification of the PV panel and the build make a big difference, especially in terms of cost [8]. Hence Levelized cost of electricity was seen as a pivotal point for choosing the different options available in the market. The given price of the battery and PV cell makes the LCOE very high, which is

unrealistic for Nepal. Hence, the following technical specification was selected based on the market survey for the PV panels and battery. These changes positively push the feasibility of decentralised energy systems.

Table 1. PV panel — JA Solar JAM54D40 N-type TOPCon (380 W variant) [8]

MATLAB variable	Old value (given)	New value (for Nepal)	Description	Status
P_PV_STC	380 W	380 W	Rated power at STC	Unchanged
G_STC	1000 W/m ²	1000 W/m ²	Standard irradiance (STC)	Unchanged
CT	-0.33e-2	-0.29e-2	Temp. coeff. of Pmax (%/°C)	Updated
T_STC	25 °C	25 °C	STC reference temperature	Unchanged
eta_PV	0.95	0.97	Module efficiency derating factor	Updated
Price	3500 €/kWh	100-120 €/kWh	Chinese supplier	[2], [3]

Table 2. Battery cell — CATL CB310 LFP prismatic (280 Ah)[9][10]

MATLAB variable	Old value	New value	Description	Status
C_cell	30×3.6 = 108 Wh	896 Wh = 280×3.2	Cell energy capacity (Wh)	Updated
eta_bat	0.95	0.98	Round-trip coulombic efficiency	Updated
SOC_min	0.50	0.20	Min. SOC limit	Changed
SOC_max	0.90	0.90	Max. SOC limit	Unchanged
Price	470 €/kWh	60 €/kWh	The Chinese supplier price for a stationary battery	[4], [5]

4.4. Flowchart of The System Design

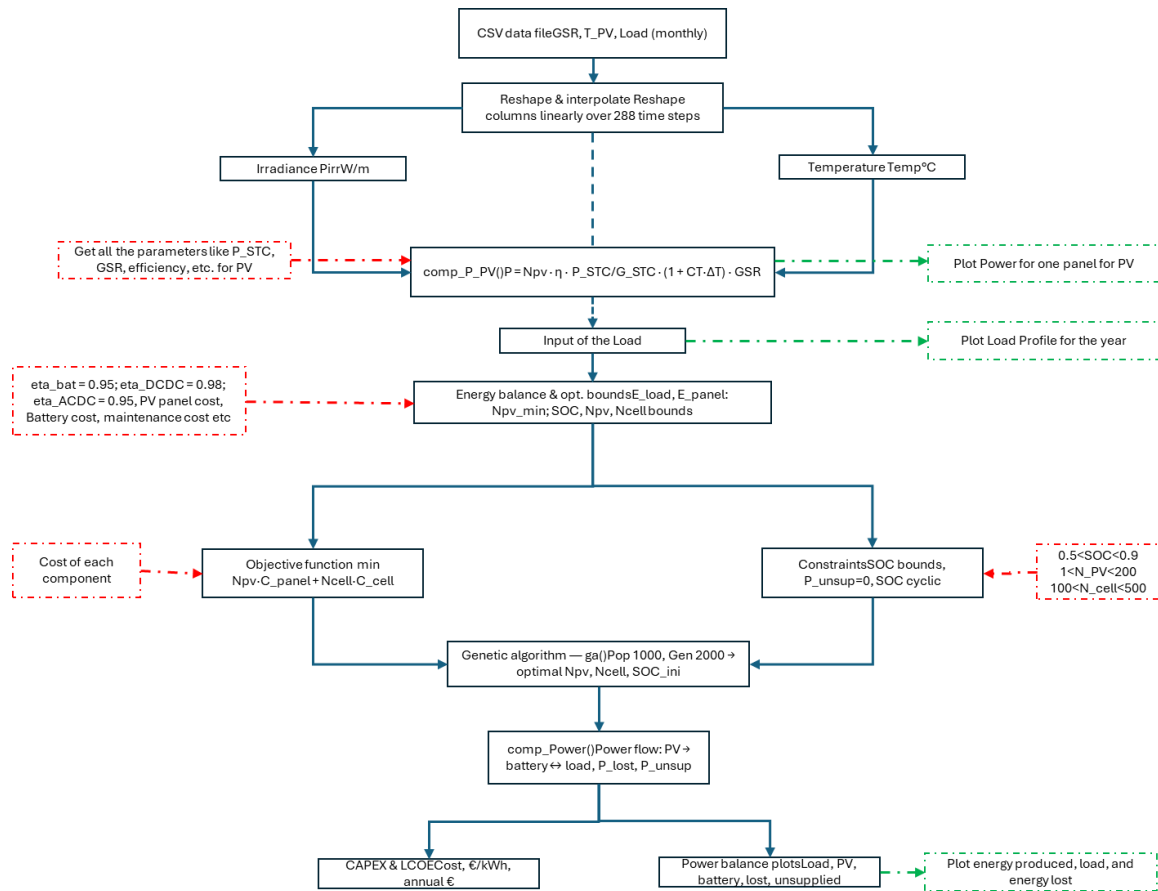


Figure 8. Proposed System Flowchart Diagram

5. Result And Discussion

5.1. Optimal System Sizing

The genetic algorithm converged to an optimal system configuration that satisfies all operational constraints while minimising capital investment. The results of the optimisation are summarised in **Error! Reference source not found.** and **Error! Reference source not found.**. It is important to note that for the base case, where the specification and price of the PV_Panel, Battery was based on a European supplier, the project was infeasible for Nepal due to the very high Levelized Cost of production. Hence, another supplier was used, which provides very low prices, and hence, as shown in **Error! Reference source not found.**[7]. The LCOE was competitive with the Nepalese grid price. It shows how the choice of technology and suppliers makes a very big difference in the feasibility of a microgrid project.

Table 3. Summary of Optimization Results for Nepal based on Given Technical Specs

Parameter	N _{pv} (panels)	N _{CELL} (cells)	SOC	CAPEX (€)	LCOE (€/kWh)
Optimal Value	82	1523	0.3144	€186,367.48	4.51

Table 4. Summary of Optimization Results for Nepal based on New Technical Specs

Parameter	N _{pv} (panels)	N _{CELL} (cells)	SOC	CAPEX (€)	LCOE (€/kWh)
Optimal Value	83	176	0.3160	€13,182.92	0.3192

The optimiser selected 82 PV panels and 1,499 CATL LFP battery cells for the given specs but for the new technical specification the optimiser selected 83 panels and 176 battery cells, with an initial SOC of 31.60%. The total capital expenditure (CAPEX) is €13,182.92, representing a substantial reduction compared to equivalent systems using the original cost parameters (€3,500/kW and €470/kWh), which would have resulted in a CAPEX approximately 30× higher. This confirms the critical importance of using up-to-date market prices in system sizing studies.

The LCOE of 0.31 €/kWh is consistent with off-grid PV-battery systems in high-altitude South Asian contexts, where transport and installation costs add to the system cost. The annual cost for the representative year is €668.04.

5.2. Total lost energy

The total loss of energy of 307.99 kWh over the representative year. This represents energy that could not be stored when the battery reached its $SOC_{max} = 0.90$ limit during periods of high PV generation and low load, primarily during summer months when irradiance peaks exceed both load and battery capacity. This energy lost is around 11% of the total energy produced over the year, which can be tapped or can be utilized by adding further load consumption facilities like hotels.

5.3. Power Balance Analysis

The power balance across all 12 representative monthly days is shown in *Figure 9*. The plot illustrates the interplay between PV generation, battery charging/discharging, load demand, lost power (surplus that cannot be stored), and unsupplied power (demand not met).

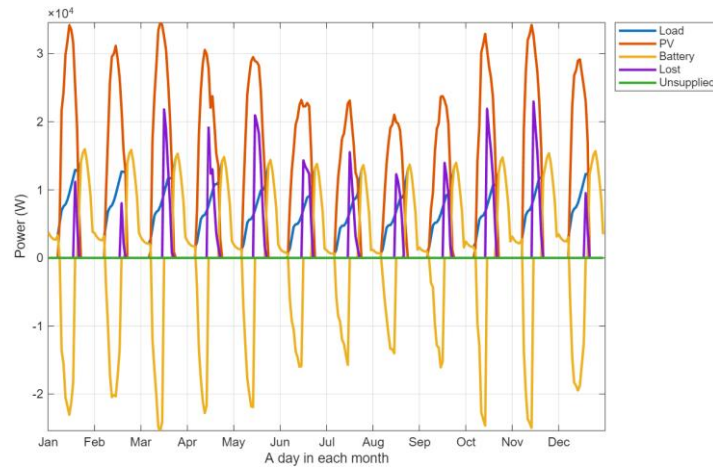


Figure 9. Annual Power Balance Load, PV Generation, Battery, Lost and Unsupplied Power (W)

Key observations from the power balance:

- **PV generation (orange):** Peaks strongly in winter months (December–January) and autumn (September–October), with moderate generation throughout the year. Generation consistently exceeds the load during daylight hours, confirming the viability of the isolated solar-battery system.
- **Battery operation (yellow):** The battery actively charges during surplus PV periods and discharges to cover nighttime and low-irradiance periods. The battery is heavily utilised throughout the year. Especially during the months of April to August the battery provides stability and meets the power demand.
- **Lost power (purple):** Appears periodically when surplus PV power exceeds both the load and the remaining battery capacity. Total lost energy of 307.99 kWh indicates some over-sizing of PV relative to storage, which could be reduced by adding more battery cells. However, in such a system, when sizing for the worst day, there will be some losses in the best days that cannot be recovered without further Capital expenditure on the storage or good export planning.
- **Unsupplied power (green):** Remains at zero throughout the entire year, confirming that the constraint of zero unsupplied load has been met. The community is fully energy self-sufficient under the optimised configuration.

Behavioural Analysis of System with Minimum SOC

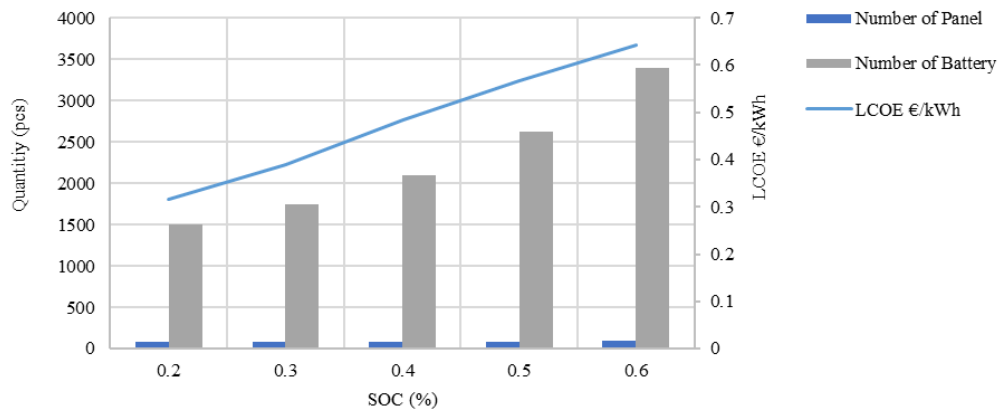


Figure 10. Effect of minimum SOC on the Cost Optimisation

The figure illustrates the impact of the initial state of charge (SOC) on system sizing and cost. As SOC increases from 0.2 to 0.6, the required number of battery cells rises significantly, while the number of PV panels shows a moderate increase. This indicates that maintaining a higher SOC requires substantial additional storage capacity to ensure energy availability throughout the operating period. Consequently, the levelized cost of electricity (LCOE) increases steadily, reflecting the higher investment associated with

battery expansion. Overall, the results highlight a trade-off between improved energy reliability and increased system cost.

5.4. Economic Analysis

Table 5. Economic Performance Summary

Metric	Value	Notes
CAPEX	€13,182.92	PV panels + battery cells
LCOE	0.31€/kWh	25-year lifetime, 8% discount rate
Annual cost (representative year)	€668.04	Based on LCOE × annual energy
Total lost energy	307.99 kWh	Surplus beyond battery capacity
Unsupplied energy	0 kWh	Full load coverage achieved
System lifetime	25 years	Per IEA 2020
Maintenance cost	2.5% of CAPEX/year	Discounted over the project life

5.5. Sensitivity Analysis

The sensitivity analysis was done to analyze the most influential parameter affecting the levelized cost of electricity production. The battery cost was found to be the most sensitive parameter (positive relation) because of the higher per-unit cost of the cells used, with a significant contribution to CAPEX. Moreover, PV cost maintenance costs are the two other major contributors to the LCOE, having a medium positive sensitivity. On the other hand, the discount rate has a negative sensitivity to the levelized cost of electricity production.

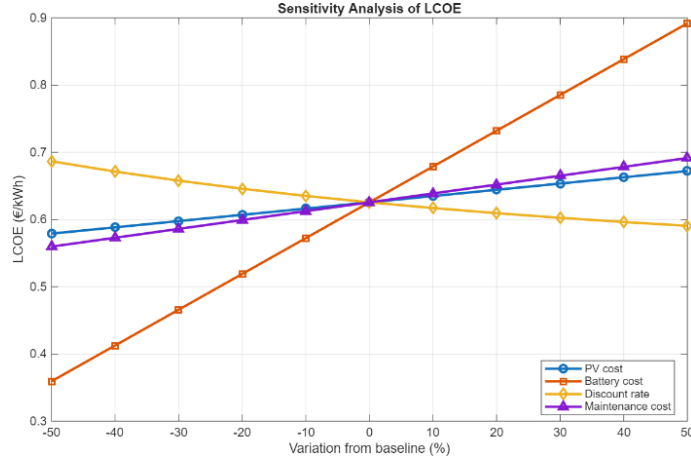


Figure 11. Sensitivity Analysis of Major Cost on LCOE

6. Conclusion

The cost optimisation was done for the village in Nepal for 50 households, incorporating around 200 people, where the supply of electricity is a problem. The load was calculated based on the reference from the literature, where the lighting load was reported to be the highest contributor to electricity use. Since the area is underdeveloped, the energy consumption per capita was limited to around 1300 kWh/year, which is considered very low worldwide. However, the geography and the higher dependency on fuel wood justify the numbers. The irradiance data were also collected from the literature, and the bounds and constraints were set for the PV panel and battery based on the technical specifications available on the internet. Then the genetic algorithm was used to optimise the cost (objective) function, i.e. capex of the battery and the PV panel, such that it satisfies all the loads and there are no blackouts.

It was found that for the project to be feasible, some cheaper alternatives to the PV panels and the battery should be determined. Hence, some Chinese suppliers were determined, who found out that for the given model, we will need 83 solar panels and 176 battery cells to completely fulfil the load, and at the same time have the lowest capital expenditure. The levelized cost of electricity was found to be 0.31 €/kWh, which is cost-competitive with the grid electricity in Nepal. However, 11% of the total generation will be lost, and it was found impossible to tap this lost power without additional expenditure on capex. This sensitivity analysis shows that the cost of the battery is the most sensitive parameter, along with the PV panel, positively affecting the LOCE.

Moreover, using the greater depth of the battery i.e., minimum SOC, provides good effects on the overall system but may lead to an unreliable system in case of an emergency, but keeping the minimum SOC requires further investment in the battery system. Hence, the decision is a trade-off between the reliability of the system and the capital investment.

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